—#— —L—e—c—t—u—r—e— —2—5—.—0—9—:— —k—N—N— ———— —K— —N—e—a—r—e—s—t— —N—e—i—g—h—b—o—u—r—s— —(—I—n—t—u—i—t—i—o—n—)— —#—#— — 📌 — —T—o—p—i—c— —O—v—e—r—v—i—e—w— —I—n— —t—h—i—s—l—e—c—t—u—r—e—,— —S—h—r—a—d—d—h—a— —m—a—'—a—m— —i—n—t—r—o—d—u—c—e—s— ————k—-—N—e—a—r—e—s—t— —N—e—i—g—h—b—o—r—s— —(—k—N—N—)————,— —a— —n—o—n—-—p—a—r—a—m—e—t—r—i—c—,— —l—a—z—y—-—l—e—a—r—n—i—n—g— —c—l—a—s—s—i—f—i—c—a—t—i—o—n— —a—l—g—o—r—i—t—h—m— —t—h—a—t— —m—a—k—e—s— —d—e—c—i—s—i—o—n—s— —b—a—s—e—d— —o—n— —g—e—o—m—e—t—r—i—c— —d—i—s—t—a—n—c—e— —a—n—d— —m—a—j—o—r—i—t—y— —v—o—t—i—n—g—.— — —-—-—-—- — —#—#— —🏛——— —A—l—g—o—r—i—t—h—m—i—c— —M—e—c—h—a—n—i—c—s— —W—h—e—n— —a— —n—e—w— —q—u—e—r—y— —d—a—t—a— —p—o—i—n—t— —a—r—r—i—v—e—s—:— —1—.— ————D—i—s—t—a—n—c—e— —C—o—m—p—u—t—a—t—i—o—n————:— —C—a—l—c—u—l—a—t—e— —d—i—s—t—a—n—c—e— —b—e—t—w—e—e—n— —t—h—e— —q—u—e—r—y— —p—o—i—n—t— —a—n—d— —e—v—e—r—y— —t—r—a—i—n—i—n—g— —p—o—i—n—t— —i—n— —t—h—e— —d—a—t—a—s—e—t—.— — — — —-— ————E—u—c—l—i—d—e—a—n— —D—i—s—t—a—n—c—e————:— —d—(—p—,— —q—)— —=— —\—s—q—r—t—{—\—s—u—m——*{—i—=—1—}—^—n— —(—p——i— —-— —q——*i—)—^—2—}——— — — — —-— —**—**M—a—n—h—a—t—t—a—n— —D—i—s—t—a—n—c—e—**—**:— —d—(—p—,— —q—)— —=— —\—s—u—m——{—i—=—1—}—^—n— —|—p——*i— —-— —q——i—|———— —2—.— ————N—e—a—r—e—s—t— —N—e—i—g—h—b—o—r—s— —S—e—l—e—c—t—i—o—n————:— —P—i—c—k— —t—h—e— ——K——— —p—o—i—n—t—s— —w—i—t—h— —t—h—e— —s—m—a—l—l—e—s—t— —d—i—s—t—a—n—c—e—.— —3—.— ————M—a—j—o—r—i—t—y— —V—o—t—i—n—g————:— —T—h—e— —m—o—s—t— —c—o—m—m—o—n— —c—l—a—s—s— —l—a—b—e—l— —a—m—o—n—g— —t—h—e— ——K——— —n—e—i—g—h—b—o—r—s— —b—e—c—o—m—e—s— —t—h—e— —p—r—e—d—i—c—t—i—o—n—.— — —-—-—-—- — —#—#— — 📊 — —V—i—s—u—a—l— —D—i—a—g—r—a—m—:— —k—N—N— —D—e—c—i—s—i—o—n— —G—e—o—m—e—t—r—y— — — – — –m–e–r–m–a–i–d– –f–l–o–w–c–h–a–r–t– –T–D– – – – – –Q–[–"–N–e–w– –Q–u–e–r–y– –P–o–i–n–t– –(–x–1–,– –x–2–)–"–]– –-–-–>– –D–[–"–C–a–l–c–u–l–a–t–e– –D–i–s–t–a–n–c–e–s– –t–o– –A–L–L– –T–r–a–i–n–i–n–g– –P–o–i–n–t–s–"–]– – – – – –D– –-–-–>– –S–[–"–S–o–r–t– –D–i–s–t–a–n–c–e–s– –&– –P–i–c–k– –T–o–p– –K– –N–e–i–g–h–b–o–r–s–"–]– – – – – –S– –-–-–>– –V–{–"–M–a–j–o–r–i–t–y– –V–o–t–e– –A–m–o–n–g– –K– –N–e–i–g–h–b–o–r–s–"–}– – – – – –V– –-–-–>–|–"–K–=–3– –(–e–.–g–.–,– –2– –O–r–a–n–g–e–,– –1– –C–y–a–n–)–"–|– –R–E–S–1–[–"–P–r–e–d–i–c–t– –O–R–A–N–G–E– –(–C–l–a–s–s– –1–)–"–]– – – – – –V– –-–-–>–|–"–K–=–5– –(–e–.–g–.–,– –2– –O–r–a–n–g–e–,– –3– –C–y–a–n–)–"–|– –R–E–S–2–[–"–P–r–e–d–i–c–t– –C–Y–A–N– –(–C–l–a–s–s– –0–)–"–]– – — – — — —-—-—-— — —#—#— —💡— —W—h—y— —C—h—o—o—s—e— —a—n— —O—d—d— ——K———?— —F—o—r— —b—i—n—a—r—y— —c—l—

a—s—s—i—f—i—c—a—t—i—o—n—,— ———a—l—w—a—y—s— —c—h—o—o—s—e— —a—n— —o—d—d— —v—a—l—u—e— —f—o—r— ——K———— —(——K—— = ——3—, ——5—, ——7—, ——\—d—o—t—s——)—.— —I—f— ——K—— —i—s— —e—v—e—n— —(—e—.—g—.—,— ——K—— = ——4——)—,— —y—o—u— —c—a—n — —g—e—t— —a— —5—0—-—5—0— —t—i—e— —(—2— —n—e—i—g—h—b—o—r—s — —f—o—r— —C—l—a—s—s— —0—,— —2— —n—e—i—g—h—b—o—r—s— —f—o—r — —C—l—a—s—s— —1—)—,— —c—r—e—a—t—i—n—g— —a—m—b—i—g—u—i—t—y — —i—n— —p—r—e—d—i—c—t—i—o—n—.— — —-—-—-— — —#—#— — 💻 — —P—y —t—h—o—n— —I—m—p—l—e—m—e—n—t—a—t—i—o—n— — — – — –p–y–t–h–o–n–

```
–i–m–p–o–r–t– –n–u–m–p–y– –a–s– –n–p– – –d–e–f– –p–r–e–d–i–c–t–_–k–n–n–_–m–a–j–o–r–i
–t–y–_–v–o–t–e–(–q–u–e–r–y–_–p–o–i–n–t–,– –d–a–t–a–s–e–t–_–p–o–i–n–t–s–,– –d–a–t–a–s
–e–t–_–l–a–b–e–l–s–,– –k–=–3–)–:– – – – – –"–"–"– – – – – –M–a–n–u–a–l– –d–e–m–o–n–s
–t–r–a–t–i–o–n– –o–f– –k–-–N–e–a–r–e–s–t– –N–e–i–g–h–b–o–r–s– –c–l–a–s–s–i–f–i–c–a–t
–i–o–n–:– – – – – –C–a–l–c–u–l–a–t–e–s– –E–u–c–l–i–d–e–a–n– –d–i–s–t–a–n–c–e–s–,– –p
–i–c–k–s– –t–h–e– –k– –n–e–a–r–e–s–t– –n–e–i–g–h–b–o–r–s–,– – – – – –a–n–d– –r–e–t–u
–r–n–s– –t–h–e– –p–l–u–r–a–l–i–t–y– –m–a–j–o–r–i–t–y– –c–l–a–s–s– –l–a–b–e–l–.– – –
– – –"–"–"– – – – – – –q–u–e–r–y– –=– –n–p–.–a–s–a–r–r–a–y–(–q–u–e–r–y–_–p–o–i–n–t–)–
– – – – –p–o–i–n–t–s– –=– –n–p–.–a–s–a–r–r–a–y–(–d–a–t–a–s–e–t–_–p–o–i–n–t–s–)– – –
– – –l–a–b–e–l–s– –=– –n–p–.–a–s–a–r–r–a–y–(–d–a–t–a–s–e–t–_–l–a–b–e–l–s–)– – – – –
– –#– –1–.– –E–u–c–l–i–d–e–a–n– –d–i–s–t–a–n–c–e–:– –s–q–r–t–(–s–u–m–(–(–x–2– –-– –x
–1–)–^–2–)–)– – – – – –d–i–s–t–a–n–c–e–s– –=– –n–p–.–s–q–r–t–(–n–p–.–s–u–m–(–(–p–o–i
–n–t–s– –-– –q–u–e–r–y–)– –*–*– –2–,– –a–x–i–s–=–1–)–)– – – – – – –#– –2–.– –F–i–n–d
– –i–n–d–i–c–e–s– –o–f– –k– –s–m–a–l–l–e–s–t– –d–i–s–t–a–n–c–e–s– – – – – –n–e–a–r–e
–s–t–_–i–n–d–i–c–e–s– –=– –n–p–.–a–r–g–s–o–r–t–(–d–i–s–t–a–n–c–e–s–)–[–:–k–]– – – –
– –n–e–a–r–e–s–t–_–l–a–b–e–l–s– –=– –l–a–b–e–l–s–[–n–e–a–r–e–s–t–_–i–n–d–i–c–e–s–]–
– – – – – –#– –3–.– –P–l–u–r–a–l–i–t–y– –m–a–j–o–r–i–t–y– –v–o–t–e– – – – – –c–o–u–n
–t–s– –=– –n–p–.–b–i–n–c–o–u–n–t–(–n–e–a–r–e–s–t–_–l–a–b–e–l–s–)– – – – – –p–r–e–d–i
–c–t–e–d–_–c–l–a–s–s– –=– –i–n–t–(–n–p–.–a–r–g–m–a–x–(–c–o–u–n–t–s–)–)– – – – – – –r
–e–t–u–r–n– –p–r–e–d–i–c–t–e–d–_–c–l–a–s–s–,– –n–e–a–r–e–s–t–_–l–a–b–e–l–s–,– –d–i–s
–t–a–n–c–e–s–[–n–e–a–r–e–s–t–_–i–n–d–i–c–e–s–]– – –#– –-–-–-– –F–u–n–c–t–i–o–n– –C–a
–l–l– –&– –T–e–s–t–i–n–g– –-–-–-– –#– –C–l–a–s–s– –0–:– –C–y–a–n– –C–r–o–s–s–e–s– –|
– –C–l–a–s–s– –1–:– –O–r–a–n–g–e– –C–i–r–c–l–e–s– –t–r–a–i–n–i–n–g–_–p–t–s– –=– –[–
[–1–,– –2–]–,– –[–2–,– –3–]–,– –[–3–,– –3–]–,– –[–6–,– –5–]–,– –[–7–,– –7–]–,– –[–8
–,– –6–]–]– –t–r–a–i–n–i–n–g–_–l–b–l– –=– –[–0–,– –0–,– –0–,– –1–,– –1–,– –1–]– –n–e
–w–_–p–o–i–n–t– – – – –=– –[–5–,– –5–]– – –p–r–e–d–,– –n–e–i–g–h–b–o–r–s–,– –d–i–s–t
–s– –=– –p–r–e–d–i–c–t–_–k–n–n–_–m–a–j–o–r–i–t–y–_–v–o–t–e–(–n–e–w–_–p–o–i–n–t–,– –t
–r–a–i–n–i–n–g–_–p–t–s–,– –t–r–a–i–n–i–n–g–_–l–b–l–,– –k–=–3–)– –p–r–i–n–t–(–f–"–Q–u
–e–r–y– –P–o–i–n–t–:– –{–n–e–w–_–p–o–i–n–t–}–"–)– –p–r–i–n–t–(–f–"–3– –N–e–a–r–e–s–t
– –N–e–i–g–h–b–o–r– –L–a–b–e–l–s–:– –{–n–e–i–g–h–b–o–r–s–}– –(–D–i–s–t–a–n–c–e–s–:–
–{–d–i–s–t–s–.–r–o–u–n–d–(–2–)–}–)–"–)– –p–r–i–n–t–(–f–"–M–a–j–o–r–i–t–y– –P–r–e–d–i
–c–t–e–d– –C–l–a–s–s–:– – –C–l–a–s–s– –{–p–r–e–d–}–"–)– – — – — — —-—-—-— — —#
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—#— — 🔑 — —K—e—y— —T—a—k—e—a—w—a—y—s— —1—.— —k—N—N— —i—s— —a— —"—l—a—z—y— —l—e—a—r—n—e—r—"—:— —i—t— —d—o—e—s— —n—o—t— —l—e—a—r—n— —e—x—p—l—i—c—i—t— —p—a—r—a—m—e—t—e—r—s— —d—u—r—i—n—g— —t—r—a—i—n—i—n—g—;— —a—l—l— —c—o—m—p—u—t—a—t—i—o—n— —o—c—c—u—r—s— —d—u—r—i—n—g— —p—r—e—d—i—c—t—i—o—n—.— —2—.— —T—h—e— —h—y—p—e—r—p—a—r—a—m—e—t—e—r— —K— —c—o—n—t—r—o—l—s— —d—e—c—i—s—i—o—n— —b—o—u—n—d—a—r—y— —s—m—o—o—t—h—n—e—s—s— —(—s—m—a—l—l— —K— —o—v—e—r—f—i—t—s—,— —l—a—r—g—e— —K— —u—n—d—e—r—f—i—t—s—)—.— —3—.— —U—s—e— —o—d—d—v—a—l—u—e—s— —o—f— —K— —i—n— —b—i—n—a—r—y— —c—l—a—s—s—i—f—i—c—a—t—i—o—n— —t—o— —p—r—e—v—e—n—t— —v—o—t—e— —t—i—e—s—.— —